

Sebastian Capellan

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EDUCATION

New York University

B.A. Data Science & Computer Science, GPA: 3.55

New York, NY

Expected May 2028

Relevant Coursework: Data Management, Operating Systems, Systems Programming, Computer Architecture, Data Structures, Data Science, Open Source Software Development, Basic Algorithms, Multivariable Calculus, Discrete Math, Linear Algebra

WORK EXPERIENCE

Duolingo

Incoming Software Engineer Intern

Pittsburgh, PA

June 2026

University of California, San Francisco

Research Assistant

Remote

October 2025 - April 2026

- Migrated a Next.js microplastics research dashboard from per-request JSON/CSV file parsing to an indexed SQLite database via custom API routes and seeding, reducing API response times by 70+%.
- Engineered a universal Python ETL pipeline that auto-infers schemas from heterogeneous JSON sources, normalizes nested objects and arrays into relational SQLite tables, and validates row counts post-migration across 10+ tables.
- Fact-checked and corrected physical property data across 18 polymer records against peer-reviewed literature and manufacturer datasheets.

PROJECTS

Quantitative Backtesting Engine | Python, SQL, DuckDB, Pandas, Pytest

- Engineered an asynchronous ETL pipelined event-driven backtester using DuckDB and yfinance, utilizing columnar data storage for 10x faster analytical queries than standard relational databases.
- Developed a vectorized crossover strategy with FIFO P&L tracking, and visualized portfolio quality with NAV curves, comparing against SPY and buy-and-hold benchmarks.
- Architected a unit testing suite with pytest for trade pairing and signal logic, preemptively resolving calculation errors.

Multithreaded Key-Value Server | C, POSIX, Ubuntu Linux

- Designed and implemented a datastore in C, supporting read-write functionality via CLI parsing and signaling.
- Engineered core functionality using POSIX system calls, enabling interaction with the operating system for file persistence, concurrency, and inter-process communication.
- Implemented datastore networking, supporting multiple clients through UNIX domain sockets and multithreading.

Cryptocurrency Portfolio Tracker | Java, Spring, MySQL

- Engineered "Coinleaf," a backend cryptocurrency portfolio tracker with RESTful API endpoints using Java, Spring Boot, and MySQL, enabling full CRUD operations and database persistence.
- Integrated live market data by consuming Coinbase API services, building a service layer to fetch and update real-time cryptocurrency prices with <200ms response times, improving accuracy of portfolio valuation.

Mandelbrot Fractal Visualizer | Python, Streamlit, Matplotlib, NumPy, Numba

- Developed and optimized fractal generation algorithm using Numba JIT compilation, NumPy vectorization, and dynamic iteration control based on zoom depth, achieving over 700% image rendering speedup.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

ColorStack @ NYU

President & Co-Founder

New York, NY

January 2025 - Present

- Foster an NYU specific chapter of an organization with 16,000+ members nationally, providing peer support, career development, and technical skill-building for the empowerment of Black and Latino students in tech.
- Spearhead outreach activity, gathering 60+ student members through targeted engagement and event hosting.

TECHNICAL SKILLS

Programming Languages: Python, SQL, C, Java, HTML, CSS

Technologies: MySQL, DuckDB, Linux, Docker, AWS, Git, GitHub

Libraries & Frameworks: Spring, NumPy, Pandas, Numba, Pytest